Jennifer Chinyere

AI/ML Engineer | Backend Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Software Engineer with experience designing and deploying intelligent platforms powered by LLMs, computer vision, OCR, and speech recognition. Built production-style AI systems using FastAPI, React, PostgreSQL, AWS, and modern cloud infrastructure, including an AI-powered clinical referral platform that reduced referral processing time from 18 minutes to 2 minutes and a computer vision system that improved radiologist throughput by 2.1×. Passionate about building reliable AI products that combine scalable backend engineering, machine learning, and intuitive user experiences to drive measurable impact.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, HTML, CSS, Bash, SQL, C++
- Frameworks: FastAPI, Django, React, Next.js, Node.js, SQLAlchemy, Pydantic, Tailwind CSS, TanStack Query
- AI/ML: OpenAI API, Groq Llama, Whisper, Tesseract OCR, Scikit-learn, MLflow, Ragas, ChromaDB, LiteLLM, YOLOv8, PyTorch, OpenCV, Grad-CAM, NumPy, Pandas, Jupyter Notebook
- Databases & Cloud: PostgreSQL, MongoDB, Redis, AWS (S3, EC2, Lambda), Render, Vercel, Railway, Supabase
- AI Developer Tools: Cursor, GitHub Copilot, Claude Code, ChatGPT, Gemini, Windsurf
- DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, REST APIs, Linux, PowerShell, Postman, Jira, Slack, Zapier
- Testing & Security: Pytest, HTTPX, Jest, Cypress, React Testing Library, JWT, OAuth2, RBAC

TECHNICAL PROJECTS

MediFlow - AI Healthcare Referral Management Platform

FastAPI, PostgreSQL, React, TypeScript, Docker, AWS, OpenAI API, Groq Llama 3.1, Whisper, Tesseract OCR, Scikit-learn(TF-IDF, Cosine Similarity), WebSockets

- Architected a multi-tenant healthcare referral platform with 50+ RESTful APIs using FastAPI, PostgreSQL, React, TypeScript, AWS, and Docker, streamlining referral intake and clinical workflows across multiple facilities.
- Built an AI-powered document intelligence pipeline using Tesseract OCR, Whisper, and Groq Llama 3.1 to automate document extraction, speech transcription, clinical summarization, and AI-assisted referral reason classification, reducing processing time from 18 minutes to 2 minutes with 92% OCR accuracy.
- Implemented duplicate patient detection using Scikit-learn and fuzzy matching techniques, alongside JWT authentication, RBAC, WebSockets, and DICOM-ready storage, contributing to a projected 45% reduction in referral handling time and approximately 360 clinical hours saved annually per clinic.

OsteoDetect - AI-Powered Bone Fracture Detection System

Python, FastAPI, PyTorch, YOLOv8, OpenCV, Grad-CAM, HTML5, CSS3, Docker, Vercel, Railway

- Engineered an AI-assisted fracture detection platform using YOLOv8, FastAPI, and OpenCV, enabling automated fracture detection from X-ray images with inference times of approximately 3 seconds per radiograph.
- Integrated Grad-CAM explainability visualizations and confidence scoring to improve clinical interpretability, contributing to an estimated 85% clinician confidence boost during evaluation.
- Built RESTful inference APIs and an end-to-end imaging workflow enabling clinicians to upload radiographs, receive AI-assisted predictions, and visualize fracture localization through an intuitive web interface.
- Optimized diagnostic workflows through AI-assisted triage, reducing manual review bottlenecks, saving an estimated 40–50 minutes per radiologist per shift, and 78% user satisfaction during testing.

EDUCATION

Diploma in Computer Science & Engineering — 2024
Akirachix — CGPA: 9.06/10